

Project Proposal

1. Project Title and Group Information

Network Layer Attack Detection using Software Defined Networking (SDN)

Group Members:

- **Muhammad Abdul Rehman** (NU ID: 23K-0609)
- **Tanzeel Adnan** (NU ID: 23K-0849)

2. Proposed Project Description (Product-Based)

This project aims to design and implement a **network-layer attack detection system** using the concept of **Software Defined Networking (SDN)**. The system will focus on identifying two common types of attacks: **Denial of Service (DoS)** and **IP spoofing**.

In traditional networks, control logic is distributed across devices, making centralized monitoring and attack detection difficult. SDN overcomes this limitation by separating the control plane from the data plane. In this project, an SDN controller will act as the central intelligence to monitor network traffic and detect abnormal behavior.

The proposed system will be developed using **Mininet** to simulate a network topology, **Ryu Controller** for centralized control, **Python** for implementing detection logic, **Scapy** for generating spoofed packets, and **Wireshark** for traffic analysis and validation.

Working of the System

1. A virtual network topology consisting of multiple hosts and a switch will be created in Mininet.
2. The Ryu controller will manage the network and receive packet information from switches.
3. Under normal conditions, hosts will communicate using standard protocols (e.g., ICMP ping), establishing baseline traffic behavior.
4. Attack scenarios will then be introduced:
 - **DoS Attack:** One host will generate a high volume of traffic targeting another host.
 - **Spoofing Attack:** Fake packets with forged IP addresses will be generated using Scapy.
5. The controller will continuously monitor incoming packets and apply detection rules:
 - For DoS: Identify unusually high packet rates from a single source.
 - For Spoofing: Detect inconsistencies such as the same IP address appearing with different MAC addresses.

6. Upon detecting suspicious activity, the system will generate alerts in the controller logs.
 7. Wireshark will be used to capture and analyze network packets to validate both normal and malicious traffic behavior.
-

Functional Features

- Simulation of SDN-based network topology using Mininet
 - Centralized traffic monitoring via Ryu Controller
 - Detection of DoS attacks using packet rate thresholding
 - Detection of IP spoofing using IP–MAC inconsistency checks
 - Generation of real-time alerts for detected attacks
 - Packet-level traffic verification using Wireshark
-

3. Plan of Work

Short-Term Plan

- Install and configure Mininet, Ryu Controller, and Wireshark
 - Create and test a simple network topology
 - Establish normal communication between hosts
 - Simulate DoS and spoofing attacks
 - Implement detection logic in the controller
 - Integrate all components and test complete workflow
 - Capture results and prepare documentation
-

Team Contribution and Work Division

To ensure efficient progress, the project will be divided into two parallel work streams:

Abdul Rehman – Controller and Detection Logic

Functionality Contributions:

- Implementation of packet monitoring within the Ryu controller
 - Development of DoS detection mechanism using packet counting and thresholding
 - Development of spoofing detection logic based on IP–MAC mapping inconsistencies
 - Generation of alert messages upon detection of suspicious traffic
 - Integration of detection logic with live network traffic
-

Tanzeel Adnan – Network Simulation and Attack Generation

Functionality Contributions:

- Design and implementation of network topology in Mininet
 - Generation of normal traffic for baseline behavior
 - Simulation of DoS attack using high-rate packet transmission
 - Implementation of spoofing attack using Scapy
 - Capturing and analyzing network traffic using Wireshark
-

4. References

1. Open Networking Foundation (ONF), *Software-Defined Networking: The New Norm for Networks*
 2. Ryu SDN Framework Documentation – <https://osrg.github.io/ryu/>
 3. Mininet Network Emulator Documentation – <http://mininet.org/>
 4. Scapy Documentation – <https://scapy.readthedocs.io/>
 5. Wireshark User Guide – <https://www.wireshark.org/docs/>
 6. Research papers on SDN-based intrusion detection systems (relevant IEEE/ACM publications)
-